

# Summary of Power and Energy Management (04/09)

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## Power Stabilization for AI Training Datacenters

### Problem and Motivation

Large AI training causes large-scale periodic power swings due to the alternation between computational phase (GPU power close to thermal design power, TDP) and communicational phase (GPU power much lower). When this phenomenon occurs at scale with thousands of GPUs at a datacenter, this becomes an infrastructure problem rather than just power management. In extreme cases, these aggregated oscillations would strain power distribution units, affect upstream transformers, and introduce harmonic frequencies that may interfere with the broader utility grid.

### Related Works

[not explicitly mentioned in the paper]

1. Large-scale distributed system about different levels of parallelism and communication frameworks

These works explain why the training is highly synchronized and scaled.

2. GPU/cloud-scale power management

These works support the solution design on the architecture level. And the combination of lower level knowledge and AI model training made this paper different from the others.

### Solution Overview

1. Software-only mitigation: Firefly

A secondary GPU workload is always injected to mitigate the power swing (usually when the primary workload enters low-power phase). It requires close monitoring on the GPU status for determining when to start and stop the secondary GPU workload. The secondary workload starts as soon as the primary workload power usage drops below a certain threshold. This keeps the power from dropping too much. At the same time, it needs to back off as soon as the primary workload starts to use more resources. By using NVIDIA's Multi-Process Service with block activity counters and matrix multiplications, Firefly was able to utilize 100% of TDP.

Challenges: 1. Performance overhead exists with Firefly since the secondary workload requires memory and compute resources; 2. This mitigation would need cooperation between end customer and cloud provider since most GPUs are accessed directly from virtual machines; 3. One fault in this process could propagate to other processes, resulting in error at scale.

Potential Optimization: 1. The failure in secondary should not affect the primary workload; 2. Prediction of communication/computation phases should be included in planning the secondary workload; 3. Scheduling of primary workload could also help arrange the secondary workload; 4. Software solution in the firmware.

## 2. GPU power smoothing

The hardware-level controls (ramp-up/down rates, minimum power floor, and stop-delay) can smooth the transitions and reduce software overhead. This might still produce power overhead and lower power utilization.

Challenge: The GPU smoothing process itself consumes power.

## 3. Energy-storage solution

The best option is rack-level energy storage, because it best solves the 4 challenges: 1. Can directly measure the load; 2. Has enough capacitance to support the workload; 3. Can meet the sudden rise/drop needs in power; 4. Can switch modes between charging and discharging quickly. This method could reduce peak power needs in training and release energy during the computation (high-power) phases.

Challenge: Having a capacitance to handle large-scale energy storage is expensive on cost, rack-level space, and embodied carbon.

## 4. Combining the three solutions above

## 5. Fast telemetry-based backstop

A telemetry-based backstop system is needed to secure the extreme oscillation which could not be smoothed by the above methods. The system could use fine-grained, low-latency telemetry and real-time spectral analysis to identify these oscillations, and initiate responses.

# Limitations

[not explicitly mentioned in the paper]

1. The solutions were evaluated mainly on simulations rather than large-scale deployment. The real-world behavior might not be as promising as the paper stated: it might depend on the task assigned, hardware, telemetry capability, etc.

2. All solutions have trade-offs.

- Secondary workload adds performance overhead
- GPU power smoothing increases power usage
- Rack-level energy storage adds cost, space, etc.

## **Future Research Directions**

Future research could focus on exploring less synchronous, more power-aware training algorithms (asynchronous training techniques, staggered scheduling, and overlap of compute and communication, etc.). These algorithms could reduce the large-scale power swing with less trade-offs.

# **AI Training Load Fluctuations at Gigawatt-scale – Risk of Power Grid Blackout?**

## **Problem and Motivation**

The main issue is that the largest AI labs are competing to build multi-gigawatt-scale datacenters, which causes huge, unpredictable fluctuations in power usage, stressing out the power grid to an unprecedented extent. The issue is twofold, the scale of these datacenters is unprecedented, and also the AI training workloads have a very unique load profile. In the worst case, this can result in a cascading collapse and blackouts for millions of Americans.

Traditional datacenter loads are pretty steady, but AI training is different. It's bursty and highly synchronized (GPUs are working in sync on the same problem). When a training run starts, reaches a checkpoint, or crashes, hundreds of thousands of GPUs can suddenly spike or drop their power draw by hundreds of megawatts in seconds.

This matters because power grids and even on-site generators aren't built to handle these kinds of rapid changes. If the load suddenly drops, you get "over-frequency" events. If it spikes, you get "under-frequency" and voltage sags. At the gigawatt scale, these fluctuations are large enough to trip power plant safety mechanisms, damage steam turbines, or cause regional grid blackouts.

## **Related Works**

[not explicitly mentioned in the paper]

The article highlights that the current industry response hasn't caught up to the reality of gigawatt-scale clusters. There currently aren't any great solutions. The current approaches include:

1. Traditional Grid Planning and Response: Utilities generally already plan for steady-state loads and sometimes use Demand Response Management Software (DRMS). However they are ill-equipped to handle fluctuations in demand in orders of gigawatts in a matter of seconds.

2. **Dummy Loads:** Some data centers currently use “load banks” (giant resistors) to burn off electricity when GPUs aren’t running, thus partially mitigating fluctuations due to sudden load drops. However, this is incredibly wasteful and generates massive amounts of excess heat.
3. **Basic Schedulers:** On the software side, datacenters often use schedulers to schedule workload, but because of economic incentives, they are typically focused on job priority and throughput rather than the electrical stability of the power grid.

## **Solution Overview**

Companies like Tesla are pushing Battery Energy Storage Systems (BESS), systems like the Megapack 2 XL, as the main hardware for dealing with the mismatch between unpredictable AI power and the inflexible power grid.

The main advantage of BESS is how quickly it can react. These systems can charge or discharge hundreds of megawatts in milliseconds or seconds, much faster or longer than alternatives like capacitor banks, or diesel generators. This allows them to smooth out the sharp “burstiness” of AI training that would otherwise trip power plant safety mechanisms or cause damage.

BESS is designed to work alongside existing generators and uninterruptible power supplies (UPS). By absorbing the initial surge of a load spike, BESS gives diesel or gas generators time to increase generation, which helps prevent their damage and extend their lifespan.

BESS also makes it possible to manage loads more flexibly. During peak hours, like a hot summer afternoon, the data center can use battery power for GPUs instead of drawing from the grid. This reduces the risk of blackouts for the utility, and the datacenter can keep running AI training at full speed.

## **Limitations**

BESS works at a technical level, but implementing it in practice and scaling it to gigawatt-sized datacenters runs into major economic and operational problems.

The upfront cost is a major barrier. Building a BESS for a gigawatt-scale datacenter can cost close to \$1 billion. This is on top of what companies building AI datacenters already spend on UPS systems and backup generators. BESS doesn’t even replace those, so it is another expensive layer.

One of the key advantages of a BESS is the demand response flexibility, allowing datacenters to run at peak capacity even during peak hours. However, this advantage is somewhat mitigated by a fundamental trade-off in State-of-Charge (SOC) management. If you discharge your battery as part of a demand response, you no longer have the energy reserved for an emergency backup event. So you have to choose between keeping the grid stable and keeping your own systems running.

Another challenge with BESS is that most utilities are far behind on IT infrastructure and their Demand Response Management Software (DRMS) is outdated. As a result it is harder for them

to predict peaks and notify customers in time to allow them to prepare, sometimes just three hours. Therefore, if the BESS isn't already charged, there isn't enough time to fill it without causing a new power spike on the grid.

Finally, the financial incentives are minimal for most markets. While BESS can certainly help stabilize the power grid, the financial returns are not large enough to justify the investment. The article notes that a 20 MW reduction might only result in a few hundred thousand dollars saved per month, which is not enough to justify a \$1 billion investment.

## **Future Research Directions**

Future Semianalysis report will dig deeper into the topic of the duplicates in ERCOT since the 108GW figure is not realistic. Besides, the future SemiAnalysis report will also explore the broader role of BESS and renewables in the power grid, in the context of a significantly higher load growth than the last 20 years.

## **Summary of Class Discussion**

[Combined two summary sections because some questions address overlapping themes from both papers.]

**Q: How would a holistic approach combining software and hardware solutions work? Would the system fall back on hardware first and then software?**

A: They are considered complementary and can be used simultaneously. For the lower level, software manages the GPU smoothing while hardware manages the energy source.

**Q: At a gigawatt scale, how much extra power or CO<sub>2</sub> emissions will be consumed to protect the grid, and is this a sustainability concern?**

A: According to the data, 10.8% extra power for Firefly. It is a fundamental trade-off between mitigating power grid instability versus the environmental sustainability cost of consuming more energy.

**Q: Could a solution like Kareus we discussed in the last lecture reduce the overall energy consumption?**

A: Kareus can be used to minimize the amplitude of the power fluctuations at the beginning, but it cannot remove the fluctuations because power stabilization and energy saving are independent of each other. If we consider resonant modes, it could actually make the grid situation worse.

**Q: What are the failure modes of these power stabilization strategies? If one of the solutions fails, how can the systems degrade gracefully?**

A: Since BESS is deployed as a parallel strategy rather than a standalone replacement for UPS or diesel generators, the failure mode is straightforward. BESS is like a bodyguard and there are also other backup systems arranged in the hardware. If BESS goes down, the system simply degrades then falls back to the existing infrastructure.

**Q: What should be the threshold for using an artificial secondary workload? Should it only be used during power emergencies or should it always be running to smooth out the power?**

A: You should always use it, even though it is not saving energy. Continuous operation is the best way to prevent dangerous power spikes. It should only be turned off when energy is cheap and grid stability is not a concern.

**Q: Can this fluctuation problem be fixed by ML algorithms rather than hardware fixes given the compute-communication phase alteration?**

A: We can mitigate the synchronization part and flatten the power profile. It is possible to use decentralized, asynchronous algorithms (like AD-PSGD) to overlap communication and execution, which effectively mitigates periodic synchronous power spikes. However, the algorithm could take a much longer time for the same results, which is a big trade-off for current technologies. But future research is promising.

**Q: Since the evaluation in these papers relies on simulation, is this a valid way to evaluate gigawatt-scale power stabilization, given the academia does not have access to such massive hardware?**

A: Researchers use tools like OpenG2G to create micro-grids connected to a few server racks. By running a micro-version of the problem, researchers can emulate traces of large LLM inference/training and evaluate how the algorithms perform even if they cannot test at full scale.

**Q: Should regulators be involved to ensure the AI data centers take on the burden of grid stability, rather than the public grid trying to pick up the slack?**

A: It would be advantageous to have a governing body managing this. Different data centers could cooperate with each other since many of them are close to each other geographically yet not work in sync. But challenges are that it is hard to execute on the company level and also requires frequent testing and maintenance.